EV Technology Course — Module 4: E-Drive Integration (Gears, E-Axles and NVH)

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The motor, inverter and gearbox used to be three separate boxes linked by cables and shafts; today they are usually one compact "e-axle", and some Chinese designs fold in the charger, DC-DC and controllers too. This module explains the single-speed gearbox, how to choose its ratio, why integration saves cost and mass, and how engineers fight the whine that an engine used to hide.

1. Learning objectives

- Explain why EVs use a single-speed reduction gear, and when two speeds make sense.
- Choose a gear ratio from top-speed and launch requirements.
- Describe 3-in-1 and multi-in-one e-drives, and disconnect units.
- Identify gear and motor noise sources and how they are reduced.
- Describe how gears, housings and e-axles are manufactured and tested.

2. Fundamentals

A motor produces torque from zero rpm and spins happily to 15,000–20,000 rpm, so a single fixed reduction of roughly 8–12:1 covers everything from launch to top speed. The gearbox only has to:

1. **Reduce speed and multiply torque** (two gear stages, or a planetary set).
2. **Split torque between the wheels** (differential).
3. **Hold the car when parked** (parking lock, since there is no engine compression or clutch).

3. Detailed technical explanation

Gearbox layouts

Layout	How it works	Pros	Cons	Examples
Parallel-axis, two-stage helical	Motor shaft → intermediate shaft → differential	Simple, efficient (~97–98%), cheap	Wider package	Most cars

Layout	How it works	Pros	Cons	Examples
Coaxial planetary	Planetary gear around the axle; motor concentric with half-shaft	Very compact	More complex, tighter tolerances	Some e-axles, many hub/2W designs
Two-speed	Adds a shiftable stage	Better launch and top speed; motor stays efficient	Cost, mass, shift quality	Porsche Taycan rear axle
Chain or belt reduction	For scooters and 3W	Cheap, quiet (belt)	Wear, adjustment	Ather 450 (belt)

Integration levels

Level	What is combined	Benefit
Separate units	Motor, inverter, gearbox linked by HV cables	Flexible sourcing
3-in-1 e-axle	Motor + inverter + gearbox in one housing	Shorter HV cables, shared cooling, ~10–20% less mass and cost
X-in-1 (e.g. 6/7/8-in-1)	Adds OBC, DC-DC, PDU, VCU, BMS controller	Even fewer connectors and housings; harder to service and to source from multiple suppliers

Disconnect units. On dual-motor AWD cars, a clutch can decouple the secondary motor at cruise to cut drag losses (or use an induction motor that drags little when unpowered).

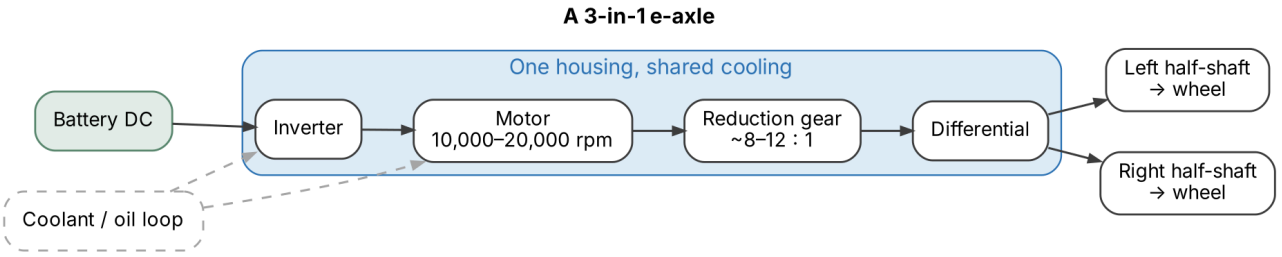
Bearings and lubrication. High motor speeds stress bearings; oil also cools the motor in many designs. EV bearings may need insulated or ceramic balls to prevent electrical discharge damage from fast-switching inverters.

NVH (noise, vibration and harshness). Without engine noise, small tonal noises become audible:

Source	Cause	Countermeasures
Gear whine	Transmission error at tooth mesh frequency	Precise grinding, micro-geometry (crowning, tip relief), helix angle, higher contact ratio

Source	Cause	Countermeasures
Motor electromagnetic whine	Torque ripple and radial forces at specific orders	Rotor skew, slot/pole choice, harmonic current injection
Inverter switching noise	PWM frequency harmonics	Higher or randomised switching frequency
Structure-borne vibration	Paths through mounts and housing	Mount tuning, housing ribs, acoustic encapsulation

4. Diagram: a 3-in-1 e-axle



5. Step-by-step: torque from motor to tyre

1. The inverter feeds the motor; the motor shaft spins at, say, 9,000 rpm with 200 N·m.
2. First gear stage reduces speed about 3:1 and triples torque.
3. Second stage reduces another ~3:1 into the differential ring gear.
4. The differential splits torque equally to both half-shafts and lets them turn at different speeds in corners.
5. Constant-velocity joints carry torque to the wheel hubs through suspension movement.
6. On regen, the path runs backwards with similar losses.

6. Engineering equations

Gear ratio from top speed:

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i_g = \frac{n_{max} \times 2\pi r_w}{60 \times v_{max}}
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Gear mesh frequency:

$$f_{mesh} = \frac{n \times z}{60}$$

Driveline efficiency and power density:

$$\eta_{drive} = \eta_{inv} \times \eta_{motor} \times \eta_{gear} \qquad \rho_P = \frac{P_{peak}}{m_{e\text{-}axle}}$$

n in rpm, z = number of teeth, v_max in m/s, ρ_P in kW/kg.

7. Worked examples

Example 1: choosing the ratio. Motor max 16,000 rpm, r_w = 0.33 m, target top speed 160 km/h (44.4 m/s): i = 16,000 × 2π × 0.33 ÷ (60 × 44.4) = **12.4**.

Example 2: what the higher ratio buys. With 250 N·m and η = 0.95, launch force at i = 12.4 is 250 × 12.4 × 0.95 ÷ 0.33 = **8,954 N**, against 6,477 N at i = 9 (Module 1). A taller ratio gives more launch force but a lower top speed; the designer balances both.

Example 3: gear whine frequency. A 20-tooth pinion at 6,000 rpm meshes at 20 × 6,000 ÷ 60 = **2,000 Hz**, right where the human ear is sensitive.

Example 4: power density. A 150 kW e-axle weighing 75 kg: **2 kW/kg**.

Example 5: driveline efficiency. Inverter 98% × motor 95% × gear 97% = **90.3%** at that operating point.

8. Real-world examples

- **BYD:** highly integrated "8-in-1" electric powertrain combining motor, gearbox, inverter, OBC, DC-DC, PDU, VCU and BMS controller.
- **Tesla:** compact drive units with integrated inverter; induction front motor in older dual-motor cars to reduce drag.
- **Porsche Taycan:** two-speed rear gearbox for launch and high-speed efficiency.
- **Suppliers:** Bosch, ZF, Schaeffler, Vitesco, GKN, Nidec, Huawei (DriveONE) and many Chinese makers sell complete e-axles.
- **India:** suppliers such as Sona Comstar (driveline gears, motors) and Bharat Forge are moving into e-drive components; many OEMs still import complete e-axles.

9. Manufacturing applications

- **Gears:** forged blanks → hobbing → heat treatment (case carburising) → **grinding/honing** (critical for NVH) → 100% gear inspection on critical features.

- **Housings:** high-pressure aluminium die casting with integrated cooling channels; machining; leak testing.
- **Assembly:** bearing press-fits, shimming for backlash, parking-lock function check, oil fill.
- **End-of-line:** run-up NVH test on a bench (order analysis), efficiency spot checks, HV and insulation tests on 3-in-1 units.
- **Costing:** integration removes cables, connectors and separate housings, but makes the unit a single high-value assembly; sourcing becomes "buy the e-axle" versus "make the gearbox, buy motor and inverter". Consider localisation content for incentives.

10. Common mistakes and misconceptions

- **"EVs don't need gears."** They need reduction and a differential; only some hub-motor designs avoid gears.
- **"More integration is always better."** It reduces cost and mass but limits supplier choice and repairability.
- **"EVs are silent, so NVH is easier."** It is harder: there is no engine noise to mask whine.
- **"Two-speed gearboxes are needed for range."** Gains are small for most cars; worth it mainly for performance or heavy vehicles.

11. Practical case study: gear whine complaints

Problem. Customers report a high-pitched whine at 60–80 km/h under light acceleration.

Analysis. Order analysis matches the first-stage mesh frequency. Measurements show transmission error higher than target on gears from one grinding machine; profile slope deviations after a wheel dressing change.

Correction. Tighten grinding parameters and inspection, introduce micro-geometry optimised for the light-load range, add 100% EOL NVH screening with an order-based limit, and improve mount isolation.

Lesson. NVH is a manufacturing-capability issue as much as a design issue.

12. Revision notes

- One fixed reduction (~8–12:1) + differential + parking lock.
- Ratio trades launch force against top speed: $i = n_{max} \times 2\pi r \div (60\, v_{max})$.
- 3-in-1 and X-in-1 integration cut mass, cost and cabling.
- Gear whine at mesh frequency; motor whine at electrical orders.

13. Glossary

Term	Meaning
e-axle / EDU	Electric drive unit: motor + inverter + gearbox
Mesh frequency	Rate at which gear teeth engage
Transmission error	Deviation from perfect motion transfer; drives whine
Micro-geometry	Tiny tooth-profile modifications for quiet running
Parking lock	Mechanical pawl that holds the car in Park
Disconnect unit	Clutch that decouples an idle motor
Order analysis	NVH analysis by multiples of shaft speed

14. Knowledge check

Conceptual

1. Why is one gear enough for most EVs?
2. What do you trade when you choose a higher reduction ratio?
3. Why is gear grinding critical for EVs?
4. Name two benefits and two drawbacks of an 8-in-1 drive.
5. Why might a dual-motor car disconnect one motor at cruise?

Numerical

1. A 25-tooth pinion at 9,000 rpm: mesh frequency?
2. Ratio needed for 140 km/h at 14,000 rpm with r_w = 0.30 m?
3. Combined efficiency of 96% motor, 98.5% inverter, 97.5% gear?

Application

1. Your OEM is deciding whether to buy complete e-axles or make the gearbox in-house. List the factors.
2. Propose a test plan to catch gear whine before vehicles ship.

15. Answer key

Conceptual

1. The motor delivers high torque from 0 rpm and works efficiently over a very wide speed range, so one ratio covers launch to top speed.
2. More launch force and gradeability against lower top speed (or higher motor speed at cruise, increasing iron losses).
3. Without engine noise, small tooth errors produce audible whine; grinding gives the precision needed.
4. Benefits: fewer cables/connectors, lower mass and cost, shared cooling. Drawbacks: single-supplier dependence, harder repair (replace large unit), thermal interactions.
5. To remove drag losses (magnet drag in a PM motor, gear churning), improving cruise efficiency.

Numerical

1. 25 × 9,000 ÷ 60 = **3,750 Hz**.
2. i = 14,000 × 2π × 0.30 ÷ (60 × 38.9) = **11.3**.
3. 0.96 × 0.985 × 0.975 = **92.2%**.

Application

1. Volume and capex (hobbing, grinding, heat treatment), in-house gear competence and NVH know-how, supplier price and margin, localisation incentives, platform commonality, IP and differentiation, warranty risk, supply security for motor magnets and semiconductors.
2. In-process gear inspection (profile, lead, pitch), transmission-error measurement on samples, 100% EOL run-up NVH with order limits, correlation of EOL data to in-vehicle acoustic tests, audit drives on vehicles from each shift.

16. Further learning

- Naunheimer et al., *Automotive Transmissions* (Springer): gear design and NVH.
- Romax, KISSsoft and GT-SUITE application notes on EV driveline NVH.
- SAE technical papers on e-axle NVH and micro-geometry.
- ORNL/DOE e-drive benchmarking.

Next: Module 5, vehicle control and regenerative braking.